

The Role of Social Capital in Selecting Interpersonal Information Sources

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Although the information-seeking literature has tended to focus upon the selection and use of inanimate objects as information sources, this research follows the more recent trend of investigating how individuals evaluate and use interpersonal information sources. By drawing from the structural, relational, and cognitive elements of social capital theory to inform antecedents to information quality and source accessibility, a research model is developed and tested. For interpersonal information sources, information quality is the key determinant of source use. Perceptions of information quality and accessibility of an interpersonal source are shown to be influenced by boundary spanning, transactive memory, and content type. Implications and prescriptions for future research are discussed.

Introduction

Organizations are social communities that facilitate the creation and transfer of knowledge (Kogut & Zander, 1996). To a large extent the field of knowledge management has arisen based on the premise that organizations should manage their information and knowledge resources in order for them to be effective (King, 2009). As important as formal knowledge management systems are for the effectiveness of retaining organizational learning, it is just as important to understand how individuals source information or knowledge from each other in the workplace (Gray & Meister, 2004). This research builds on a growing body of work looking at how individuals choose the interpersonal sources they use (Woudstra & Van den Hooff, 2008; Xu, Kim, & Kankanhalli, 2011; Yuan, Monge, & Fulk, 2005).

At its most basic level, searching for information is a response to a need (Case, 2012). An individual chooses to go

to an external source when they perceive that they lack necessary information or experience (Levitt & March, 1988). Within organizational settings, information or knowledge can reside in any number of different sources, including a variety of documents, systems, or people. Some organizational knowledge is even embedded within organizational policies or culture (Alavi & Leidner, 2001; Grant, 1991; Nonaka & Takeuchi, 1995). With these resources being so widely distributed throughout the organization, a major challenge for information and knowledge management is getting the necessary resources to individuals when they are needed (Huber, 1991). This makes it important to understand the factors that influence how individuals make decisions in choosing the sources use.

Knowledge and information are often conceptualized as related but distinct concepts. Information is typically defined as contextualized data whereas knowledge is seen as more personal and the result of interpretation and experience (Nonaka & Takeuchi, 1995). From an organizational perspective the management of both information and knowledge is often subsumed under knowledge management. As noted by Grover and Davenport (2001), “what companies actually manage under the banner of KM is a mix of knowledge, information, and unrefined data” (p. 7). Although the differences between information and knowledge are often debated, in this research we do not explicitly differentiate between information seeking and knowledge sourcing. This study focuses on the factors that influence the use of interpersonal sources, so it deals with the sourcing of both information and knowledge. At different times an individual going to an interpersonal source might be seeking information or knowledge depending on the situation. What they receive as the result of the interaction could likewise be either information or knowledge. This research sees information and knowledge seeking both as part of a broader process in which seekers select the sources they will use (Gray & Durcikova, 2006).

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Prior source selection work has viewed selecting a source as the end result of two different processes. The first view argues that source selection is the end result of cost–benefit analysis. According to this view, the accessibility (or cost) of a source is compared to the quality (or benefit) of the source, and a source will be selected if the benefits outweigh the costs (Allen, 1984; O'Reilly, 1982; Zimmer, Henry, & Butler, 2008). The second view, rooted in media richness theory and social presence theory, purports that selection stems from a drive to reduce uncertainty (Daft, Lengel, & Trevino, 1987; Short, Williams, & Christie, 1976). When compared, the cost–benefit view has been the more prevalent source selection paradigm followed by information seekers (Carlson & Davis, 1998).

Although prior work has established the efficacy of the cost–benefit view, there is less understanding of what factors influence the cost–benefit calculus in source use (Agarwal, Xu, & Poo, 2011). In conceptualizing sources as either relational or nonrelational (human or nonhuman), research has found that individuals applied the cost–benefit calculus differently depending on the type of source (Zimmer et al., 2008). In particular accessibility was not a factor in the selection of human sources even though it was for nonhuman sources. Other studies have found that the factors that increase the use of sourcing from colleagues are different from those of documents and electronic repositories (Gray & Durcikova, 2006).

Although there is a variety of sources available to individuals, prior research has shown that individuals often have a preference for interpersonal information sources (Allen, 1984; Fidell & Green, 2004; Hertzum, 2002). Woudstra, van den Hooff, and Schouten (2012) have argued that aspects of social capital form the underlying dimensions of perceptions of accessibility for interpersonal sources. Rather than conceptualizing social capital as the dimensions of accessibility, this research looks at the impact of the dimensions of social capital on perceptions of accessibility and quality for interpersonal sources. The results of this analysis provide insights into how individuals evaluate interpersonal information sources, particularly in the organizational settings where social capital is formed. This research seeks to deepen our understanding of the factors that encourage the use interpersonal sources.

This study investigates the role of social capital in the sourcing of information or knowledge from interpersonal information sources. The following section develops the research model and formally presents the study hypotheses. Following the presentation of the research model the methodology is described and the results are presented. The article concludes with a discussion of the implications of the findings and directions for future research.

Research Model and Hypotheses

This research looks at the evaluation and use of interpersonal information sources to accomplish work-related tasks. Interpersonal information sources have also been

referred to as relational sources (Zimmer et al., 2008) and human sources (Woudstra & Van den Hooff, 2008). Because prior literature has reliably demonstrated that the choice to use a particular source is predominantly determined via cost–benefit analysis (Carlson & Davis, 1998), the constructs quality and accessibility are used to operationalize perceptions of costs and benefits of these sources. According to social capital theory, the relationships an individual has within his social network facilitates the conditions necessary for information exchange to occur (Nahapiet & Ghoshal, 1998). As a result the lens of social capital has increasingly been used to explore the use of interpersonal information sources (Woudstra & Van den Hooff, 2008; Woudstra et al., 2012; Xu et al., 2011). This article argues that the network of relationships an individual has with others should influence his perceptions concerning the relative costs and benefits associated with going to others as a source of information. The constructs of boundary spanning, content type, and transactive memory are used to operationalize the structural, cognitive, and relational dimensions of social capital. The proposed research model is shown in Figure 1 and its related hypotheses developed in the following sections.

Social Capital

An organization can be viewed as a community that creates and transfers knowledge (Kogut & Zander, 1996); the knowledge created and held is unique to the individual, and not the organization (Grant, 1996). All the members of a network do not have equal access to the same resources, including knowledge. Because of the underlying assumption that knowledge is unique to the individual, a social capital perspective, where individuals differ with respect to their level of social capital, is an ideal lens for investigating how individuals use interpersonal sources.

According to Nahapiet and Ghoshal (1998), social capital is “the sum of the actual and potential resources embedded within, available through, and derived from the network of relationships possessed by an individual or social unit” (p. 243). These authors define the three interrelated dimensions of social capital: structural, relational, and cognitive. The *structural* dimension concerns the linkages between individuals. It comprises the overall connections between network actors, including both who is contacted and how they are contacted (Burt, 1992). The *relational* dimension represents the relationships individuals form with one another via repeated interactions (Granovetter, 1992). This includes the knowledge assets created via interpersonal interaction. The *cognitive* dimension refers to the creation of shared meaning that develops as individuals interact with one another.

Information and knowledge seeking can be seen an inherently social process, particularly when investigating interpersonal search patterns (Xu, Kim, & Kankanhalli, 2010). Nahapiet and Ghoshal (1998) develop their theory of social capital arguing that the three dimensions of social capital

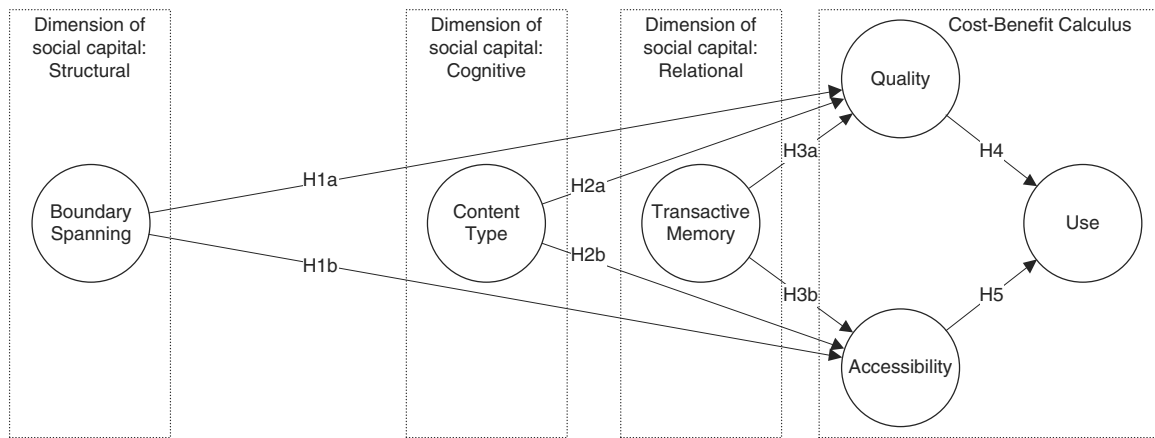


FIG. 1. Proposed research model.

facilitate the development of knowledge by enabling information exchange and combination to occur. Information exchange and combination consist of access to the other party, anticipation of the value of the exchange, motivation to exchange, and compatibility in the content of the exchange. In an organizational environment, it can be assumed that by attempting to consult an interpersonal information source there is sufficient motivation. When attempting to consult interpersonal sources, seekers make themselves vulnerable by displaying gaps in their own knowledge (Morrison, 1993), hence by crossing the cognitive threshold of going to search, motivation can be assumed. This leaves accessing the other party (cost) and the anticipation of a valuable exchange (benefit) as the two major drivers through which social capital impacts source selection. We argue that social capital will encourage the use of interpersonal sources by increasing the perceived accessibility and quality of these sources.

Boundary Spanning

Individuals who demonstrate high levels of communication across organizational boundaries impact how information or knowledge is shared and combined (Ancona & Caldwell, 1992). These “boundary spanners” are individuals who create connections and they serve an important role as information processors (Aldrich & Herker, 1977). Most commonly boundary spanners are thought of as operating where the organization meets its external environment, but as organizations have grown more complex, boundary spanners also bridge boundaries within an organization (Pawłowski & Robey, 2004). They are responsible for developing contacts with sources across boundaries and relaying what they learn so that others can better cope with the external environment. By virtue of boundary spanners, a network of individuals is tied together by information relatedness. In other words, structural embeddedness is a

necessary condition for boundary spanners to exist. Individuals seek out others and are sought out by others as sources of job related information and knowledge creating bridges that link individuals within the organization. These activities create the connections at the heart of organizational interaction. They have advantages over others that do not have these connections because they fill “structural holes” (Burt, 1992). As a result, boundary spanning represents the ties that define the structural dimension of social capital.

Seen from this perspective, boundary spanning is a role an individual takes on or develops over time through interacting and developing relationships with interpersonal sources, which, in turn, gives structure to the existing social network. As organizations become more specialized, internally idiosyncratic language and jargon naturally develop within a group. Boundary spanners have the ability to understand and recode that specialized language, developing shared meaning for the interacting partners, which makes the information or knowledge they acquire usable by others (Tushman & Scanlon, 1981). Through boundary spanning, an individual is in a better position to make assessments of the quality of the content from interpersonal sources. This suggests the following hypothesis:

H1a: Individuals performing more boundary spanning activities will perceive the quality of interpersonal information sources to be higher.

Individuals who act as boundary spanners have easier access to sources because of their more central network position (Burt, 1992). Boundary spanners also have established avenues for sharing with others. This more centralized location should make it easier for boundary spanners to access others within the network. This suggests the following hypothesis:

H1b: Individuals performing more boundary spanning activities will perceive the accessibility of interpersonal information sources to be higher.

Information and knowledge are often seen as separate, but they can also be seen as points on a continuum capturing differences in experience and interpretation. The content that individuals seek in organizational settings ranges along this continuum from that which is easily transferred (explicit) to that which is only transferred with great difficulty (tacit) (Polanyi, 1966). Although many categorizations exist, most of them incorporate ease (or lack thereof) of transfer into their conceptualization. For example, Byström and Järvelin (1995) define domain information, which consists of theories, laws and other easily codified information, and problem solving information which is defined as expertise in treating a given problem. This distinction reflects the difference between tacit and explicit content.

Our conceptualization of content type represents this continuum and corresponds to the cognitive dimension of social capital. Shared experiences, narratives, and meaning provide the contextualization that is used to create new knowledge (Nahapiet & Ghoshal, 1998). As the content that individuals seek moves from more explicit to more tacit it requires more interpretation to be understood. Because individuals seek information in response to an assessment that their current understanding is insufficient for the task at hand, it is useful to consider the extent to which a source can convey more tacit content in evaluating the impact of social capital.

What makes tacit content difficult to transfer also makes it hard to gauge its quality. Explicit knowledge, which is easily transferred, should be much more readily evaluated with respect to its quality. This is because explicit knowledge is easily codified and transferred, which allows the searcher to more readily evaluate it objectively.

Because of the ease with which explicit information can be transferred, it is expected that this distinction will influence perceptions of accessibility. Shared meaning and narratives can more easily be created when dealing with more explicit content. This content should be more widespread in the network the individual seeker is accessing. It is reasonable to expect that easily codified content would be available in more locations than tacit content. This means that a searching individual has the opportunity to find explicit information in potentially more information sources without having to create a specific shared meaning and understanding with another individual. Because of its lower entropy, the likelihood of information being in more places, and the ease with which explicit content can be transferred we suggest the following hypotheses:

H2a: Individuals who search for more explicit content will perceive the quality of interpersonal information sources to be higher.

H2b: Individuals who search for more explicit content will perceive the accessibility of interpersonal information sources to be higher.

Transactive memory is a mechanism through which groups collectively encode, store, and retrieve knowledge (Wegner, Erber, & Raymond, 1991). It represents an understanding of “who knows what” that develops over time in close relationships. The concept was originally developed to explain the behavior of couples, but the concept has since been applied to understanding interactions in groups and organizations (Moreland & Myaskovsky, 2001). Individuals in continuing relationships come to rely on others to supplement their own limited cognitive resources (Mohammad & Dumville, 2001). Borgatti and Cross (2003) refer to knowing what another person knows as an “enduring relational characteristic” that facilitates knowledge and information sharing. As noted by Lewis (2003) “whereas transactive memory exists in the mind of an individual, a transactive memory system exists between individuals as a function of their transactive memories” (p. 588).

Transactive memory systems allow individuals to become more specialized or differentiated within the groups in which they exist (Lewis, 2003; Wegner et al., 1991). Members of the group develop “implicit structures for assigning responsibility for information based on their shared conception of one another’s expertise” (Brandon & Hollingshead, 2004, p. 633). Individuals become reliant on one another to be responsible for certain expertise. This mutual reliance establishes the norms, obligations, and expectations that reflect the relational dimension of social capital.

As groups acquire new information or knowledge it is categorized with regards to a local expert within the transactive memory system (Rulke-Liang & Rau, 2000). The members of the system then relegate the formal storing of the information to the local expert (Nevo & Wand, 2005). This implies that the individuals within the network have ongoing relationships with one another, which keeps the knowledge within the network. This way a member only has to know the general topic of the information, and who the local expert is. From an information quality standpoint, this maximizes the quality of the information because the best source is in charge of storing the information. This suggests the following hypothesis:

H3a: There will be a positive relationship between transactive memory and the perceived information quality of interpersonal information sources.

The value of a transactive memory system is in the value of the relationships the individuals have with one another, and the increased access to information those relationships enable (Yuan et al., 2005). By having well-developed transactive memory, an individual is likely to view interpersonal sources as more easily accessible in comparison to an individual lacking transactive memory. This suggests the following hypothesis:

H3b: There will be a positive relationship between transactive memory and the perceived accessibility of interpersonal information sources.

Source Use

Information quality has been the focus of many research efforts (Arazy, Nov, Patterson, & Yeo, 2011; DeLone & McLean, 1992; Nelson, Todd, & Wixom, 2005). Despite this, no consistent definition of information quality exists. Quality can refer to the information retrieved from a system, or it can refer to the system itself (DeLone & McLean, 1992). Additionally, quality can be viewed as subjective (i.e., the information meets expectation [Hilligoss & Rieh, 2008]), or quality can be viewed objectively (i.e., the information meets the requirements of a particular task [Taylor, 1986]). For this work, a general definition of “fitness for use” is adopted (Ge & Helfert, 2007), which incorporates both the objective and subjective view of information quality (Wang & Strong, 1996).

Information quality is typically viewed as a benefit in source selection research. Prior work involving information quality argues that it is a multidimensional construct (McKinney, Yoon, & Zahedi, 2002; Woudstra et al., 2012; Zmud, 1978). For the purposes of this research, the focus is solely upon information quality and not information system quality (DeLone & McLean, 1992). Remembering that individuals search for information to address shortcomings in their own personal knowledge base, focusing on “fitness for use” as a definition of quality is appropriate. In this context high quality information refers to its usefulness for the task at hand without the need for additional clarification or refinement.

The literature demonstrates that individuals prefer to use sources that are understandable, reliable, and accurate (O'Reilly, 1982). Some studies have made the claim that quality is the main driver in the use interpersonal information sources (Woudstra et al., 2012; Zimmer et al., 2008). Additionally, individuals show a deeper level of trust in high quality information (Menon & Varadarajan, 1992; Moriarty & Speckman, 1984). This suggests the following hypothesis:

H4: There will be a positive relationship between perceived quality and the use of interpersonal information sources.

Accessibility has been argued to be the most influential factor in source selection, with individuals forgoing higher quality sources for more easily accessible lower quality ones (Culnan, 1983, 1985; Gerstenberger & Allen, 1968; O'Reilly, 1982). Accessibility has been treated as a multidimensional construct (Culnan, 1983, 1984, 1985; Woudstra et al., 2012; Zmud, Lind, & Young, 1990). Culnan (1983, 1985) argued that accessibility can be viewed as either terminal or informational. Terminal accessibility focuses on physical access to a source. In modern organizations, physical access is no longer the issue it once was. The advent of new information communication technologies (IM, e-mail, text, voice mail, etc.) makes accessing another individual much easier. Informational accessibility refers to accessing the information once an individual has accessed the source. In an interpersonal setting this means how effectively the source can transfer information to the seeker.

Because use is predicated on being able to access the information source, greater perceived accessibility of a source will result in a greater likelihood of an individual consulting that source. This suggests the following hypothesis:

H5: There will be a positive relationship between perceived accessibility and the use of interpersonal information sources.

Research Methods

A survey was used to collect the data to test the research model. The data were collected from part time MBA students, all employed by various organizations. Because the focus is on information seeking in response to work-related tasks, respondents were instructed to think of their information-seeking behaviors for typical work tasks.

Measures

Whenever possible, items used to operationalize constructs were taken from the existing scales. In some cases slight changes were made to fit the context, although these changes were minimal. The first concern is to measure use, accessibility, and quality for information sources. Four interpersonal sources individuals routinely use to accomplish work tasks were identified. These sources are listed and defined in Table 1, and cover the most commonly used types of interpersonal sources an individual might consult while in the workplace. In subsequent analyses information quality and source accessibility are treated as second-order factors. Use, however is treated as a first-order formative factor.

Accessibility for each source was measured using four semantic differential items from Zmud et al. (1990). Quality has been conceptualized at both the system and informational level. Only items designed to measure information quality were used. Six Likert-scaled items to measure informational quality across three dimensions (understandability, reliability, and usefulness) were used (McKinney et al., 2002). Like accessibility, quality was measured for each source type. Transactive memory was measured using four Likert-scaled items from Peltokorpi (2004). These items reflect relational social capital. For example one of the items reads “I know who is assigned to what task in my company.”

TABLE 1. Listing of interpersonal information source types.

Source	Examples
Supervisor	Individual one reports to
Coworkers inside department	Those within one's own organizational department
Coworkers outside department	Those within the organization, but outside one's own department
Others outside the organization	Professional contacts outside the organization

Knowing this information about a coworker implies some degree of prior relationship with the individual in question. The degree to which an individual acts as a boundary spanner was measured using two Likert-scaled items from Keller and Holland (1975). These items reflect structural social capital by capturing information about others seeking information from them, and their seeking information from others. Transactive memory and boundary spanning are personal characteristics and are only measured once for each respondent. Source use was measured using items created by Zimmer et al. (2008) and the scale was comprised of four items. Finally, information type was measured using a three item semantic differential scale from Zimmer and Henry (2007). This represents the cognitive dimension of social capital because of the understanding between interacting partners that develops when sharing complex knowledge. All items are shown in Appendix B.

Data Collection

Data were collected from MBA students at two large Southeastern universities in the United States. Only respondents who were currently working in information intensive occupations were allowed to participate. The use of part-time evening students also provides a sample from an array of organizations that makes it reasonable to generalize the findings more broadly to a working professional. All respondents were assured their responses would remain anonymous and they did not have to provide their names. Of the approximately 500 students invited to participate, 204 met the employment criterion and chose to participate representing a response rate of 40.8%.

Results and Analysis

In this section, we present general demographic information about the sample followed by the tests of our hypotheses.

Demographics

Demographic information about the respondents is presented in Table 2. As can be seen, most respondents were in their 20s and befitting a sample predominantly in that age range, most were in the beginnings of their career having been with their particular organization for 5 years or less. Additionally, most worked for a small- or medium-sized organization, though a third of respondents worked for firms with over 1,000 individuals. Finally almost half of the respondents identified themselves as managers whereas the rest reported that they were technically oriented (e.g., programmer, analyst, systems administrator).

Data Analysis

Partial least squares path modeling (PLS-PM) was used to test the research hypotheses. PLS-PM is a structural

TABLE 2. Participant demographics.

Gender			Career Orientation		
	n	%		n	%
Male	123	60.3	Managerial	99	48.5
Female	79	39.1	Technical	80	39.2
Age			Size of the Organization		
	n	%		n	%
20–24	38	19.3	under 100	46	23.7
25–29	80	40.6	100–499	60	30.9
30–39	57	28.9	500–999	22	11.3
40–49	19	9.7	1,000–4,999	34	17.5
50 +	3	1.5	5,000 +	32	16.5
Tenure with the Organization					
			n	%	
under 1 year			13	6.5	
1 year			37	18.4	
2 years			36	17.9	
3 years			20	10	
4 years			21	10.5	
5 years			21	10.5	
6–10 years			36	17.7	
over 10 years			17	8.5	

equation modeling technique that tests both the structural and measurement models simultaneously (Bollen, 1989). PLS-PM is appropriate and preferred over other SEM techniques (e.g., LISREL) when the research focus is on theory building (Chwelos, Benbasat, & Dexter, 2001). Additionally, PLS-PM has less stringent sample size requirements than a technique such as LISREL, and does not assume multivariate normality (Qureshi & Compeau, 2009). The following analyses were performed using SmartPLS (version 2.0) (Ringle, Wende, & Will, 2005).

Measurement Model

Before the hypotheses were formally tested, the reliability and validity of the model were assessed. Reliability was evaluated via the average variance extracted (AVE) and the composite reliability (CR). The AVE measures item variance relative to measurement error, and should be in excess of 0.50 in order to justify using the construct (Barclay, Thompson, & Higgins, 1995; Fornell & Larcker, 1981). The CR is interpreted much like Cronbach's α and a value in excess of 0.70 is considered acceptable for research (Fornell & Larcker, 1981). As can be seen in Table 3, all study constructs display adequate CRs and AVEs.

Evaluating convergent and discriminant validity was performed in three ways. First, the correlations among the constructs were compared to the square root of the AVE. If the square root of the AVE is greater than the correlations, discriminant validity is demonstrated (Chin, 1998). The

TABLE 3. Reliabilities and correlations of study constructs.

	AVE	CR	(1)	(2)	(3)	(4)	(5)
Accessibility (1)	0.88	0.92	0.94				
Boundary Spanning (2)	0.81	0.79	0.20	0.90			
Content Type (3)	0.78	0.73	0.01	0.35	0.88		
Quality (4)	0.87	0.94	0.51	0.24	-0.06	0.93	
Transactive Memory (5)	0.77	0.87	0.23	0.15	-0.01	0.40	0.88
Use*	—	—	0.26	0.25	0.02	0.45	0.19

Note. *Use is measured as a formative construct therefore AVE and CR is not calculated.

Root AVE is bolded on the diagonal.

square root of the AVE is shown on the diagonal in Table 3, and these values exceed all the off diagonal values. The second method to assess validity is to examine the pattern of factor loadings. Each item should load more highly on its own factor than the other factors (Chin, 1998). Based on an exploratory factor analysis (EFA) with a varimax rotation, all items load on their own constructs with no evidence of any cross loading (Tabachnick & Fidell, 2001). The results of the EFA are shown in Table 4 of Appendix A. The final way to assess validity is to look at the item loadings on their respective factors. Each item should significantly load on its respective factor. Every item for every construct save one loaded significantly on its factor. The largest p value for the items that significantly loaded on their factor was 0.006. The item loads are shown in Table 5 of Appendix A. One of the information type items did not significantly load on the information type factor and its p value was 0.15.

All subsequent analyses were conducted twice. Once with the nonsignificant information type item included and once with it dropped. None of the statistical analyses were impacted by including or dropping the item. Based on that, it was decided to leave the information type scale with three items instead of dropping it to two items. All items included on the survey can be seen in Appendix B. Satisfied as to the reliability and validity of the study constructs; the next section provides the results of the hypothesis testing.

Structural Model

PLS-PM estimates path coefficients and then a bootstrapping procedure is used to calculate the t -statistics, standard errors, and p -values for the hypothesis tests (Chin, 1998). The path coefficients are interpreted as those from a multiple regression, and the R^2 is much the same, an indication of the amount of explained variance in the dependent construct (Barclay et al., 1995). The path coefficients and significance tests are shown in Figure 2.

Hypothesis 1a tests the relationship between boundary spanning and quality, specifically that boundary spanners are better able to assess the quality of the information that is being considered for use. Hypothesis 1a was supported. Boundary spanning displayed a positive relationship with

quality, $\beta = 0.24$, $t = 3.16$, $p = 0.0008$, meaning that boundary spanners assess the information quality of information sources to be higher.

Hypothesis 1b tests the relationship between boundary spanning and accessibility, specifically that boundary spanners will rate interpersonal information sources as more accessible. Hypothesis 1b was supported. Boundary spanning displayed a positive relationship with accessibility, $\beta = 0.19$, $t = 3.71$, $p = 0.0001$, meaning that boundary spanners view human information sources are more accessible.

Hypothesis 2a tests the relationship between content type and quality, specifically that when content that is more closely associated with knowledge on the data→information→knowledge continuum is required, seekers will have a more difficult time assessing its quality. Hypothesis 2a was supported. Information type displayed a negative relationship with information quality, $\beta = 0.14$, $t = 1.78$, $p = 0.0375$, meaning that individuals seeking more explicit content evaluated interpersonal sources to be of higher quality.

Hypothesis 2b tests the relationship between content type and accessibility, specifically that more explicit sources will be more accessible than tacit sources. Hypothesis 2b was not supported, $\beta = 0.05$, $t = 0.79$, $p = 0.2148$. One possible explanation for this finding is that when considering an array of sources, the type of information does not enter into the decision calculus. Rather than considering if an interpersonal information source can provide the information in question, all that the seeker considers is whether the source is available. Future research into the decision calculus individuals use in choosing sources is required in order to better understand why this relationship was not supported.

Hypothesis 3a tests the relationship between transactive memory and quality. Hypothesis 3a was supported. Transactive memory displayed a positive relationship with quality, $\beta = 0.36$, $t = 4.42$, $p < 0.0001$, meaning that individuals with well-developed transactive memories evaluate the quality of an interpersonal information source to be higher.

Hypothesis 3b tests the relationship between transactive memory and accessibility, specifically that individuals with a well-developed transactive memory will view interpersonal sources as more accessible. Hypothesis 3b was supported. Transactive memory displayed a positive relationship with accessibility, $\beta = 0.20$, $t = 4.96$, $p < 0.0001$; meaning that those with well-developed transactive memories view interpersonal sources as more readily accessible.

Hypothesis 4 tests the relationship between quality and interpersonal source use. Confirming previous research (O'Reilly, 1982, 1983; Zimmer et al., 2008), Hypothesis 4 was supported. Quality was positively related to source use, $\beta = 0.44$, $t = 5.75$, $p < 0.0001$, suggesting that individuals prefer to use higher quality interpersonal information sources.

Hypothesis 5 tests the relationship between accessibility and source selection. Older research often found support for the relationship between accessibility and source use (Gerstenberger & Allen, 1968; Hardy, 1982; O'Reilly, 1982), however more recent research has seen this

TABLE 4. EFA results for study constructs.

	transactive memory	boundary spanning	quality: supervisor	quality: department coworkers	quality: coworkers	quality: others	accessibility: supervisor	accessibility: departmental coworkers	accessibility: coworkers	accessibility: others	content type
TM1	0.68	0.231	0.269	0.166	0.044	-0.064	0.004	0.104	0.098	-0.147	0.094
TM2	0.847	0.144	0.155	0.118	-0.028	0.023	-0.022	0.011	0.138	0.047	0.078
TM3	0.812	-0.06	-0.036	0.155	0.087	0.024	0.163	0.035	0.07	-0.017	-0.089
TM4	0.642	-0.13	0.28	0.181	0.11	0.082	0.058	0.058	-0.104	0.048	-0.03
BS1	0.09	0.761	0.094	0.059	0.131	0.081	-0.006	0.107	0.053	0.107	0.275
BS2	0.026	0.807	0.038	0.051	0.027	0.073	-0.055	0.061	0.003	0.141	0.066
Q:SU1	-0.022	0.03	0.813	0.176	0.052	-0.057	0.247	-0.064	0.128	-0.026	-0.022
Q:SU2	-0.023	0.151	0.736	0.169	0.083	0.177	0.259	-0.048	0.13	-0.044	-0.026
Q:SU3	0.197	-0.073	0.835	0.199	0.102	-0.078	0.12	0.073	-0.018	0.045	-0.057
Q:SU4	0.232	-0.053	0.847	0.146	0.108	-0.097	0.089	0.087	-0.027	0.007	-0.04
Q:SU5	0.11	0.086	0.881	0.159	0.096	-0.049	0.1	0.069	0.057	-0.01	-0.03
Q:SU6	0.108	0.048	0.859	0.124	0.137	-0.068	0.155	0.091	-0.028	-0.075	-0.036
Q:ID1	0.17	0.149	0.139	0.795	0.093	0.116	0.105	0.098	0.108	0.071	-0.082
Q:ID2	0.158	0.109	0.067	0.759	0.197	0.045	0.079	0.121	0.117	0.043	-0.168
Q:ID3	0.084	-0.002	0.211	0.849	0.139	0.122	0.033	0.135	0.097	0.05	0.066
Q:ID4	0.107	0.034	0.19	0.848	0.183	0.084	0.038	0.182	0.032	0.104	0.066
Q:ID5	0.142	-0.081	0.233	0.851	0.162	0.037	-0.053	0.155	0.045	0.084	0.049
Q:ID6	0.051	-0.029	0.164	0.816	0.248	0.048	0.008	0.195	0.008	0.047	0.062
Q:CO1	0.097	0.051	0.094	0.211	0.851	0.115	0.099	-0.022	0.175	0.059	-0.016
Q:CO2	0.003	0.093	-0.011	0.213	0.83	0.152	0.083	0.007	0.169	0.122	-0.041
Q:CO3	0.061	-0.008	0.14	0.148	0.856	0.188	-0.033	0.087	0.181	0.017	-0.012
Q:CO4	0.019	-0.004	0.1	0.114	0.871	0.182	-0.066	0.096	0.144	0.06	0.031
Q:CO5	0.027	0.043	0.148	0.158	0.868	0.187	-0.014	0.057	0.098	0.097	-0.021
Q:CO6	0.043	0.047	0.107	0.138	0.851	0.23	-0.04	0.091	0.12	0.038	0.006
Q:O1	0.128	0.123	-0.069	0.089	0.165	0.834	0.019	-0.084	0.01	0.217	-0.039
Q:O2	0.056	0.134	-0.045	0.061	0.183	0.809	0.049	-0.117	0.03	0.202	0.037
Q:O3	-0.051	-0.007	-0.012	0.037	0.127	0.905	-0.089	0.057	0.024	0.13	0.009
Q:O4	-0.046	-0.031	-0.004	0.082	0.161	0.905	-0.089	0.076	0.02	0.175	0.021
Q:O5	-0.016	0.039	-0.007	0.072	0.153	0.883	-0.054	0.055	-0.001	0.195	-0.012
Q:O6	0.024	-0.043	-0.051	0.07	0.2	0.88	-0.042	0.036	0.013	0.154	0.022
A:SU1	0.036	-0.028	0.244	-0.019	-0.002	-0.06	0.859	0.187	0.127	0.055	-0.026
A:SU2	0.092	-0.028	0.47	0.014	-0.011	0.041	0.653	0.297	0.07	0	-0.008
A:SU3	0.106	-0.028	0.248	0.1	0.046	-0.117	0.814	0.249	0.123	0.015	-0.005
A:SU4	0.034	-0.01	0.247	0.092	0.005	-0.072	0.861	0.208	0.112	0.015	0.038
A:ID1	0.057	0.151	0.044	0.177	0.067	-0.04	0.292	0.806	0.212	0.085	-0.004
A:ID2	-0.014	-0.041	0.125	0.42	0.058	0.093	0.161	0.719	0.218	0.082	-0.006
A:ID3	0.09	0.043	0.016	0.203	0.093	0.016	0.275	0.838	0.181	0.055	0.045
A:ID4	0.08	0.087	0.084	0.282	0.118	0.003	0.206	0.818	0.192	0.082	-0.001
A:CO1	0.016	0.118	0.115	0.043	0.226	-0.055	0.102	0.199	0.822	0.206	-0.01
A:CO2	-0.029	-0.057	0.083	0.166	0.341	0.058	0.14	0.21	0.697	0.202	-0.016
A:CO3	0.049	-0.008	0.019	0.124	0.24	0.043	0.108	0.174	0.851	0.173	0.054
A:CO4	0.075	0.016	0.021	0.083	0.185	0.063	0.115	0.171	0.881	0.195	0.037
A:O1	0.026	0.142	-0.036	0.052	0.12	0.331	0.039	0.111	0.231	0.771	0.012
A:O2	0.018	0.18	0.026	0.141	0.064	0.369	0.056	0.097	0.115	0.778	0.030
A:O3	-0.016	-0.043	-0.056	0.082	0.136	0.346	0.003	0.047	0.256	0.803	-0.022
A:O4	0.043	0.016	-0.003	0.107	0.084	0.334	0.004	0.046	0.225	0.822	-0.013
INFO1	0.004	0.19	-0.128	0.008	-0.024	0.067	0.102	0.057	-0.019	-0.067	0.743
INFO2	0.027	-0.102	0.088	0.025	-0.086	-0.007	-0.12	-0.046	0.085	-0.067	0.769
INFO3	-0.012	0.238	-0.133	-0.052	0.075	-0.033	0.024	0.013	-0.033	-0.016	0.791

TABLE 5. Factor item loadings.

	transactive memory	boundary spanning	quality: supervisor	quality: department coworkers	quality: coworkers	quality: others	accessibility: supervisor	accessibility: departmental coworkers	accessibility: coworkers	accessibility: others	content type	use
TM1	0.773											
TM2	0.841											
TM3	0.779											
TM4	0.763											
BS1		0.92										
BS2		0.782										
Q:SU1			0.861									
Q:SU2			0.806									
Q:SU3			0.889									
Q:SU4			0.884									
Q:SU5			0.915									
Q:SU6			0.9									
Q:ID1				0.846								
Q:ID2				0.816								
Q:ID3				0.909								
Q:ID4				0.924								
Q:ID5				0.915								
Q:ID6				0.886								
Q:CO1					0.908							
Q:CO2					0.886							
Q:CO3					0.917							
Q:CO4					0.913							
Q:CO5					0.925							
Q:CO6					0.907							
Q:O1						0.893						
Q:O2						0.865						
Q:O3						0.911						
Q:O4						0.934						
Q:O5						0.922						
Q:O6						0.915						
A:SU1							0.917					
A:SU2							0.837					
A:SU3							0.924					
A:SU4							0.931					
A:ID1								0.909				
A:ID2								0.868				
A:ID3								0.932				
A:ID4								0.935				
A:CO1									0.912			
A:CO2									0.856			
A:CO3									0.932			
A:CO4									0.943			
A:O1										0.906		
A:O2											2.226	
A:O3											816	
A:O4											688	
INFO1											658	
INFO2											8	
INFO3											87	
USE1											0	869
USE2											0	202
USE3											0	807
USE4											0	0.344

Value marked with *p<.05 does not significantly load on its latent factor.

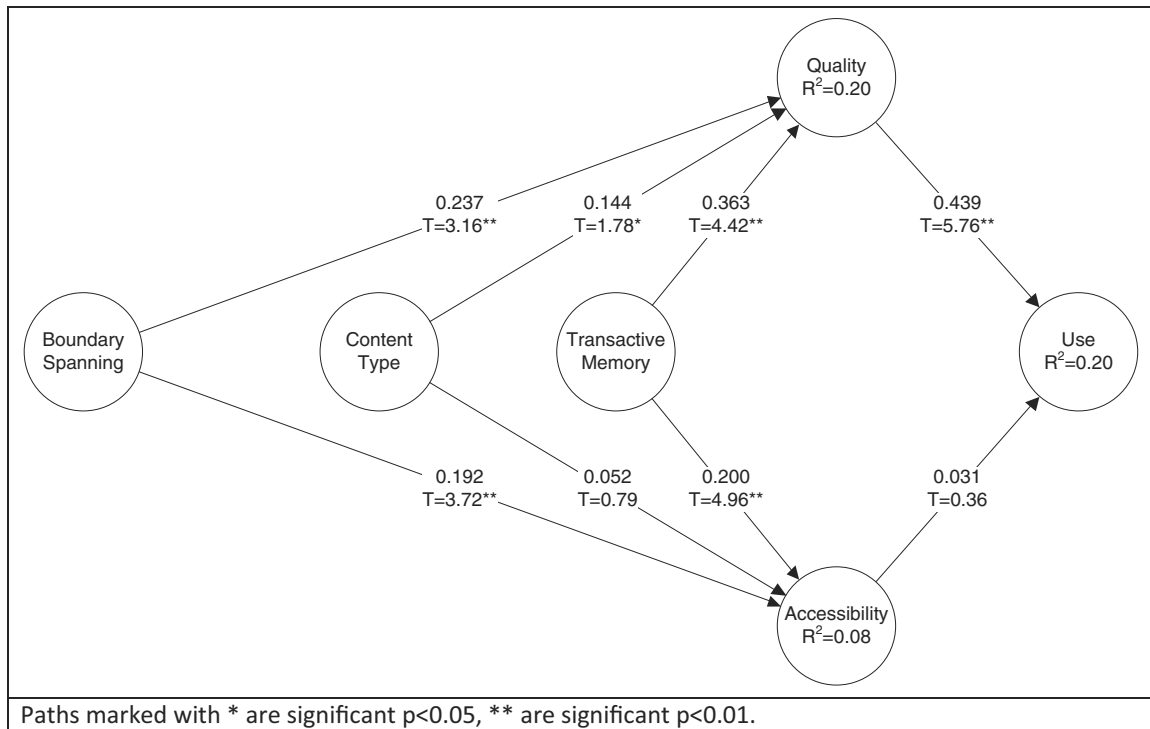


FIG. 2. Path coefficients and significance results.

relationship go unsupported (Xu, Tan, & Yang, 2006; Zimmer et al., 2008). As with these more recent studies, this relationship was not supported, $\beta = 0.03$, $t = 0.36$, $p = 0.36$. A possible explanation that reconciles these findings lies in the definition of “information sources.” The older studies focused on nonhuman information sources such as books, magazines, and trade journals. The more recent studies, including this one, focused on interpersonal information sources. An avenue for future research could be to explicitly compare the differences in how information seekers’ search strategies change when consulting human sources versus consulting nonhuman sources. Early research into this area demonstrates that there are some differences (Zimmer et al., 2008), however more work still needs to be done.

Another possible explanation for these findings involves how information communication technologies (ICTs) have influenced how individuals find and exchange information. Early studies found that individuals preferred human information sources (Allen, 1984; Gerstenberger & Allen, 1968). At the time, interpersonal communication was largely limited to face-to-face contact or telephone contact. Today ICTs have revolutionized how individuals can interact. Future research should investigate how individuals leverage these technologies to acquire and share information.

Discussion

This study explored the effects of social capital on the use of interpersonal sources. Although quality and accessibility

have long been theorized to impact source selection (Culnan, 1985; O’Reilly, 1982), the role of accessibility has demonstrated mixed results (Allen, 1984; Gerstenberger & Allen, 1968; Woudstra et al., 2012; Zimmer et al., 2008). These results, focusing exclusively on interpersonal information sources, support recent findings in which the role of accessibility has been called into question (Woudstra et al., 2012; Zimmer et al., 2008). Whether accessibility being a nonissue is the result of the increased availability of interpersonal sources because of the impact of communication technologies is an avenue future research should explore. This is a critical distinction between having the ability to access another source versus choosing to access another source. Several major process models of information search, all in some way or another, address this choice process in formulating which source to further investigate (Bates, 1989; Ellis, 1989; Fidell & Soergel, 1983; Kuhlthau, 1991; Vakkari, 2001). Hence, if a source is not on that list, it has no chance of being selected for use. Future research should further investigate the impact of accessibility and its impact on the decision calculus of selecting potential sources.

Unlike accessibility, quality did display a significant relationship with source selection. These results suggest that for interpersonal sources use is primarily based on perceptions of quality. This supports previous work in the information search domain (O’Reilly, 1982; Vakkari, 1999; Woudstra et al., 2012; Zimmer et al., 2008). It should also come as a relief to source designers and organizational managers that individuals base their decision to use an interpersonal

information source on quality instead of satisficing between quality and accessibility in their decision making. These findings could suggest that individuals leverage their interpersonal network and then select individuals who they feel will provide the best quality information instead of trading that off against how easily they can get to another.

An important caveat of these findings is they apply only to interpersonal information sources. Future research needs to extend these findings. One of the premises of a computer as a social actor has been to make interacting with technology more like interacting with another individual (Moon, 2000). Can impersonal information sources be designed in such a way that individuals perceive of them no differently than interpersonal sources? If that is indeed the case, then the next question is, do individuals source information from these “hybrid” sources like they do from purely interpersonal sources? These are two interesting avenues for future research.

Social Capital, Quality, and Accessibility

This study investigated the three dimensions of social capital and their impact on perceptions of information quality and source accessibility. As discussed, boundary spanning corresponds to the structural dimension of social capital, content type corresponds to the cognitive dimension, and transactive memory corresponds to the relational dimension. This conceptualization differs from previously published work (e.g., Woudstra et al., 2012), by focusing on social capital as an antecedent of perceptions of quality and accessibility. By developing a better understanding of how individuals determine the quality and accessibility of interpersonal information sources, organizations can better manage and utilize their human capital, and by extension help social capital better develop within the organization.

As expected, boundary spanning displayed a significant relationship with both quality and accessibility. A boundary spanner is an individual who has developed significant ties with others, both within and between organizations. These ties allow for the sharing of information across organizational boundaries. The structural and cognitive aspects of boundary spanning—having the network ties and having developed a shared language with others—make interpersonal information sources more accessible. By the same token, those network ties and shared experiences also make it easier for an information seeker to judge the quality of the information that is being shared. Boundary spanners engage in several different roles, ambassadors and task coordinators in particular are roles that require ongoing relations with significant others to acquire information, resources, and support to address current requirements (Ancona & Caldwell, 1992; Faraj & Yan, 2009). These relationships that boundary spanners have with other individuals fulfill the relational dimension of social capital. Norms and obligations may also develop over time that help boundary spanners assess the quality of the information being provided.

Content type, which corresponds to the cognitive dimension of social capital, displayed mixed results. With regards to quality, information type was negatively related to quality, meaning that perceived quality was lower for more tacit content. Knowledge that can be easily codified and transferred should not only be more easily judged in terms of quality, but the process through which the knowledge was created more commonly understood. These explicit narratives and experiences facilitate information transfer, whereas tacit knowledge remains more difficult to transfer. Content type displayed no relationship with accessibility. Nahapiet and Ghoshal (1998) argue that when individuals have a shared background it should make them more readily accessible to one another. However we do not find the cognitive dimension's impact on accessibility. Organizational members typically have a shared history and narratives with one another. It may be that those from outside this shared history are not viewed as “inaccessible” but rather as “non-existent.” Whether or not this is included in the decision calculus to access a particular source should be investigated in future research.

Another explanation for this finding could be a lack of statistical power. To investigate this possibility, the method for post hoc power analysis described in MacCallum, Browne, and Sugawara (1996) was conducted for this hypothesis. This analysis provides a minimum value for power, and the power of this analysis was 0.69. Extrapolating power for this analysis from the published tables in Reinartz, Haenlein, and Henseler (2009) places the power for this test at 0.74. Although both of these numbers are less than the 0.80 power level (Cohen, 1988), it is believed that this analysis had sufficient power, and that the nonsignificant result observed was not because of a lack of power. Future research should be conducted to substantiate this lack of finding.

Transactive memory may act as both structural and relational aspects of social capital, and it also performed as expected. Individuals with well-developed transactive memory systems reported that the interpersonal sources they used were of higher quality and were more accessible than those without such a mechanism in place. With regards to quality, members of the transactive system know which other members are the experts in the area where information is needed. Because expertise is an indicator of quality (Palazzolo, 2005), it makes sense that there was a relationship between an individual's transactive memory and his assessment of the quality of a relational source. As for accessibility, knowing who knows something is half the battle in finding information.

One area this study does not address is impact of the strength of the ties among network participants selecting interpersonal sources. Previous research has shown that strong ties are more effective in transferring tacit knowledge (Krackhardt, 1992; Maurer, Bartsch, & Ebers, 2011; Reagans & McEvily, 2003), whereas weak ties are more effective in transferring explicit knowledge (Hansen, Podolny, & Pfeffer, 2001). Future research should investigate how the strength of the ties among network

participants influences the assessment of the costs and benefits of using an interpersonal information source.

We have operationalized the structural, relational, and cognitive dimensions of social capital using particular measures that capture specific aspects of these dimensions. As noted by Nahapiet and Ghoshal (1998), "Although we separate the three dimensions analytically, we recognize that many features we describe are, in fact, highly interrelated" (p. 243). As a result, other studies that choose to capture different aspects of the dimensions of social capital may find different results. This work should be seen as part of a larger narrative on the role that social capital plays in how individuals share information and knowledge within organizations.

Implications for Practitioners

This research also has several prescriptions for practitioners. Whereas prior works have shown that selection is based on a tradeoff between quality and accessibility (Carlson & Davis, 1998; Zimmer et al., 2008), this study extends these findings and provides key insights for managers responsible for providing information sources.

First, practitioners need to be aware of how social capital dimensions impact perceptions of costs and benefits of interpersonal sources. Managers should be cognizant of ways to further enable and enhance tools that enhance dimensions such as transactive memory systems (Sparrow, Liu, & Wegner, 2011; Wegner, 1995). This will allow for individuals to develop important social contacts and increase their access to higher quality, well-vetted information. Additionally, although transactive memory has traditionally been portrayed as a system that develops among interacting individuals (Hollingshead, 1988, 1998a, 1998b, 2001), technology has allowed individuals to expand their transactive memory beyond interpersonal sources. Practitioners should also work to leverage the strengths of these transactive memory systems. This includes making sure the content is valid, but also being cognizant of the social aspects involved when individuals interact with these information communication technologies (Xu et al., 2011).

Second, it is important to recognize that not all sources are created equal. As accessibility becomes less important in interpersonal information sources because of the pervasiveness of technology, individuals will select sources based increasingly on quality and the suitability of that source and the quality of the information contained to the task at hand (Agarwal et al., 2011). Practitioners need to assess the types of tasks being performed to make sure the best people are made available as information sources. Failure to do so can lead to individuals working inefficiently (Xu et al., 2011).

Conclusion

In a world where attention is a limiting factor when it comes to individual information processing, it is important to understand where individuals go for the information they

require. This study contributes to our understanding of interpersonal source use by looking at the impact of social capital dimensions. We find that the social context in which individuals find themselves impacts the assessments of both the quality and accessibility of interpersonal sources. Understanding this allows individuals to better leverage social capital and organizations to make better use of their knowledge resources.

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Appendix A

This appendix shows additional statistical results. Table 4 shows the EFA results. All items load strongly on their respective factor with very little amounts of cross loading. Table 5 shows the item loads on its particular factor. All items load significantly on their respective factors with the exception of the INFO2 item. That item does not load significantly on the information type latent construct. Use is a formative construct hence it is not included in the EFA results.

	Transactive memory	Boundary spanning	Quality: supervisor	Quality: department coworkers	Quality: coworkers	Quality: others	Accessibility: supervisor	Accessibility: departmental coworkers	Accessibility: coworkers	Accessibility: others	Content type
TM1	.680	.231	.269	.166	.044	-.064	.004	.104	.098	-.147	.094
TM2	.847	.144	.155	.118	-.028	.023	-.022	.011	.138	.047	.078
TM3	.812	-.060	-.036	.155	.087	.024	.163	.035	-.017	.070	-.089
TM4	.642	-.130	.280	.181	.110	.082	.058	.058	-.104	.048	-.030
BS1	.090	.761	.094	.059	.131	.081	-.006	.107	.053	.046	.275
BS2	.026	.807	.038	.051	.027	.073	-.055	.061	.003	.141	.066
Q:SU1	-.022	.030	.813	.176	.052	-.057	.247	-.064	.128	-.026	-.022
Q:SU2	-.023	.151	.736	.169	.083	.177	.259	-.048	.130	-.044	-.026
Q:SU3	.197	-.073	.835	.199	.102	-.078	.120	.073	-.018	.045	-.057
Q:SU4	.232	-.053	.847	.146	.108	-.097	.089	.087	-.027	.007	-.040
Q:SU5	.110	.086	.881	.159	.096	-.049	.100	.069	.057	-.010	-.030
Q:SU6	.108	.048	.859	.124	.137	-.068	.155	.091	-.028	-.075	-.036
Q:ID1	.170	.149	.139	.795	.093	.116	.105	.098	.108	.071	-.082
Q:ID2	.158	.109	.067	.759	.197	.045	.079	.121	.117	.043	-.168
Q:ID3	.084	-.002	.211	.849	.139	.122	.033	.135	.097	.028	.050
Q:ID4	.107	.034	.190	.848	.183	.084	.038	.182	.032	.104	.066
Q:ID5	.142	-.081	.233	.851	.162	.037	-.053	.155	.045	.084	.049
Q:ID6	.051	-.029	.164	.816	.248	.048	.008	.195	.008	.047	.062

	Transactive memory	Boundary spanning	Quality: supervisor	Quality: department coworkers	Quality: coworkers	Quality: others	Accessibility: supervisor	Accessibility: departmental coworkers	Accessibility: coworkers	Accessibility: others	Content type
Q:CO1	.097	.051	.094	.211	.851	.115	.099	-.022	.175	.059	-.016
Q:CO2	.003	.093	-.011	.213	.830	.152	.130	.007	.169	.122	-.041
Q:CO3	.061	-.008	.140	.148	.856	.188	-.033	.087	.181	.017	-.012
Q:CO4	.019	-.004	.100	.114	.871	.182	-.066	.096	.144	.060	.031
Q:CO5	.027	.043	.148	.158	.868	.187	-.014	.057	.098	.097	-.021
Q:CO6	.043	.047	.107	.138	.851	.230	-.040	.091	.120	.038	.006
Q:O1	.128	.123	-.069	.089	.165	.834	.019	-.084	.010	.217	-.039
Q:O2	.056	.134	-.045	.061	.183	.809	.049	-.117	.030	.202	.037
Q:O3	-.051	-.007	-.012	.037	.127	.905	-.089	.057	.024	.130	.009
Q:O4	-.046	-.031	-.004	.082	.161	.905	-.089	.076	.020	.175	.021
Q:O5	-.016	.039	-.007	.072	.153	.883	-.054	.055	-.001	.195	-.012
Q:O6	.024	-.043	-.051	.070	.200	.880	-.042	.036	.013	.154	.022
A:SU1	.036	-.028	.244	-.019	-.002	-.060	.859	.187	.127	.055	-.026
A:SU2	.092	-.028	.470	.014	-.011	.041	.653	.297	.070	.000	-.008
A:SU3	.106	-.028	.248	.100	.046	-.117	.814	.249	.123	.015	-.005
A:SU4	.034	-.010	.247	.092	.005	-.072	.861	.208	.112	.015	.038
A:ID1	.057	.151	.044	.177	.067	-.040	.292	.806	.212	.085	-.004
A:ID2	-.014	-.041	.125	.420	.058	.093	.161	.719	.218	.082	-.006
A:ID3	.090	.043	.016	.203	.093	.016	.275	.838	.181	.055	.045
A:ID4	.080	.087	.084	.282	.118	.003	.206	.818	.192	.082	-.001
A:CO1	.016	.118	.115	.043	.226	-.055	.102	.199	.822	.206	-.010
A:CO2	-.029	-.057	.083	.166	.341	.058	.140	.210	.697	.202	-.016
A:CO3	.049	-.008	.019	.124	.240	.043	.108	.174	.851	.173	.054
A:CO4	.075	.016	.021	.083	.185	.063	.115	.171	.881	.195	.037

	Transactive memory	Boundary spanning	Quality: supervisor	Quality: department coworkers	Quality: coworkers	Quality: others	Accessibility: supervisor	Accessibility: departmental coworkers	Accessibility: coworkers	Accessibility: others	Content type	Use
TM1	0.773											
TM2	0.841											
TM3	0.779											
TM4	0.763											
BS1		0.920										
BS2		0.782										
Q:SU1			0.861									
Q:SU2			0.806									
Q:SU3			0.889									
Q:SU4			0.884									
Q:SU5			0.915									
Q:SU6			0.900									
Q:ID1				0.846								
Q:ID2				0.816								
Q:ID3				0.909								
Q:ID4				0.924								
Q:ID5				0.915								
Q:ID6				0.886								
Q:CO1					0.908							
Q:CO2					0.886							
Q:CO3					0.917							
Q:CO4					0.913							
Q:CO5					0.925							
Q:CO6					0.907							
Q:O1						0.893						
Q:O2						0.865						
Q:O3						0.911						
Q:O4						0.934						
Q:O5						0.922						
Q:O6						0.915						

Appendix B

Shown here are all the survey items.

Transactive memory taken from Peltokorpi (2004). Anchors: *strongly disagree–strongly agree*.

1. I am able to identify a person with relevant knowledge in my company.
2. I know who has what kind of specialized expertise in my company.
3. I know who is assigned to what task in my company.
4. People in my work unit know the right people and sources of information to perform efficiently.

Boundary spanning taken from Keller and Holland (1975). Anchors: *strongly disagree–strongly agree*.

1. Fellow employees frequently seek job related information from me.
2. Individuals who are not part of my organization frequently seek job related information from me.

Quality items taken from McKinney et al. (2002). Anchors: *strongly disagree–strongly agree*. Each of the four different sources was measured with all six items.

1. The information I get from <source> is <six different items>.
 - a. clear in meaning
 - b. easy to comprehend
 - c. accurate
 - d. credible
 - e. informative
 - f. valuable

Accessibility items were taken from Culnan (1984). Semantic differential scale with adjective pairs shown here. Each of the four different sources was measured with all four items.

1. To me, <source> are:
 - a. available/ unavailable
 - b. dependable/ undependable
 - c. convenient/ inconvenient
 - d. accessible/ inaccessible

Content type items were developed by Zimmer and Henry (2007). Semantic differential scale with adjective pairs shown here.

Indicate the type of content you generally search for:

1. explicit/ tacit
2. easy to express/ difficult to express
3. specific/ vague

Interpersonal source use items were developed by Zimmer and Henry (2007). Anchors: *very infrequently*–*very frequently*
How often do you use <source> in your daily work tasks

- a. Your supervisor
- b. Coworkers in your department
- c. Coworkers not in your department
- d. Others outside your organization